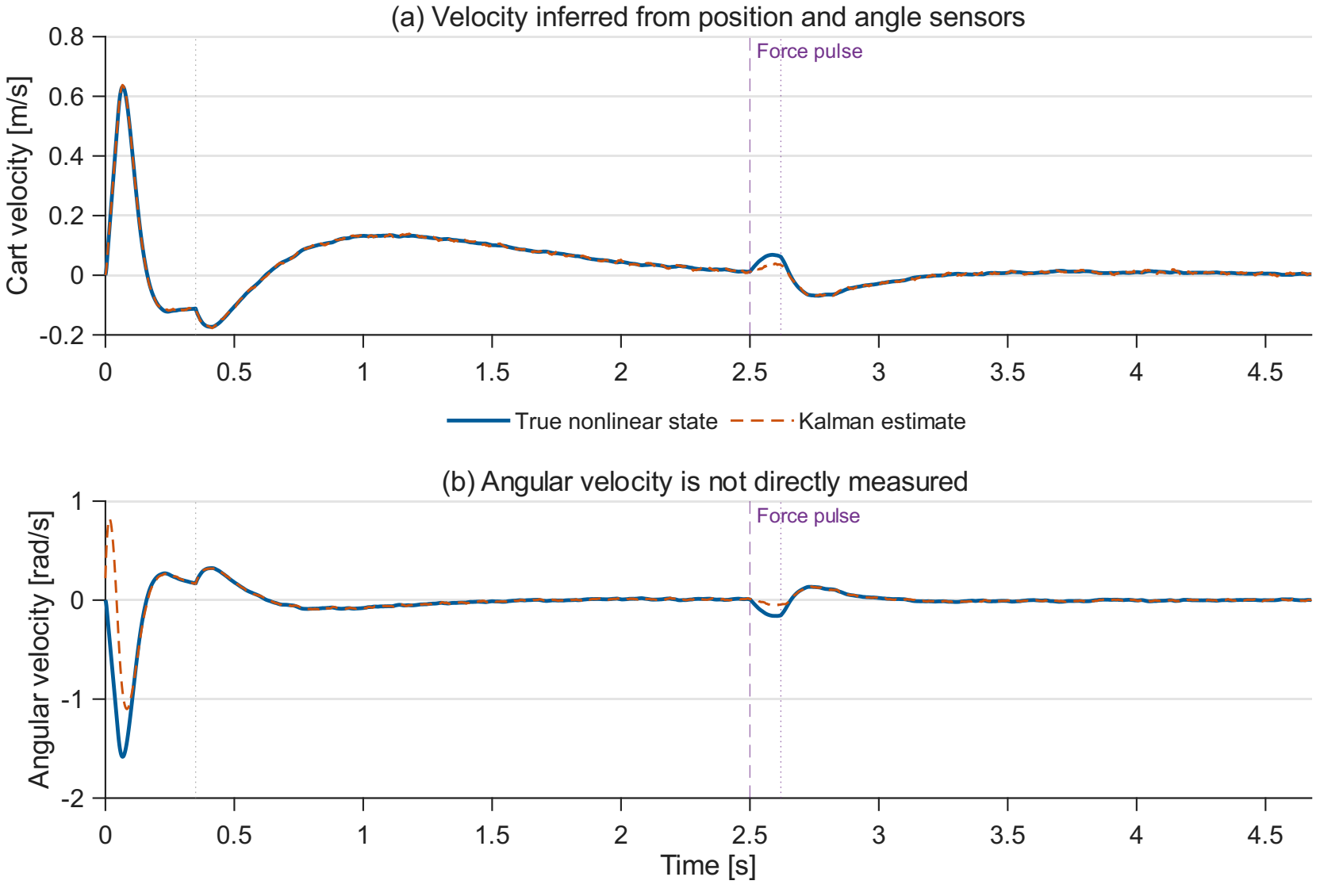

(a) Velocity inferred from position and angle sensors
Force pulse
Cart velocity [m/s]
0.8
0.6
0.4
0.2
0
-0.2
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
True nonlinear state
Kalman estimate
(b) Angular velocity is not directly measured
Force pulse
Angular velocity [rad/s]
1
0
-1
-2
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time [s]